



LINAC Access Control Interlock System Validation and Verification Report – Oct 2008

1.5.52.5 Rev. 0

Date: 2008-October-22

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REVISION HISTORY

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A	2008-Oct-16	Issued For Review	Lyndon Cowles
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1.0 PURPOSE

This report outlines the results from Validation and Verification (V&V) of the LINAC Access Control Interlock System (ACIS) conducted from September 30 – October 02, 2008.

2.0 BACKGROUND

The validation and verification of the LINAC ACIS ensures that it is operating within the design specifications outlined in “1.2.37.2.Rev.0-Personnel Safety Lockup System-West [1] and in the 1.2.37.3.Rev.0-Personnel Safety Lockup System Micro84 Program-West [2]. The V&V procedure also verifies the “LINAC-LTB1 Lockup Procedure” 2.7.37.2 Rev.2 [3].

3.0 ACRONYMS

ACIS	Access Control Interlock System
BST	Beam Stop
EOS	Emergency Off Station
EXB	Exit Button
HRN	Horn
HSE	Health Safety and Environment Department
HV	High Voltage
H/W	Hard Wired
LOTO	Lock Out Tag Out
MKS	MKS Integrity Client Problem Tracking System
RCC	Radiological Control Coordinator
V&V	Validation and Verification
ZLL	Zone Lockup Lights

4.0 VERIFICATION AND VALIDATION RESULTS

4.1 SUMMARY

Two non-conformances were identified during the V&V testing and were entered into the MKS problem tracking system.

1) MKS 4612

During the annual V&V testing of the LINAC, the following claxon was not working when tested.

Zone 4 nearest to LUS 9

The claxon was tested and was working but it had been turned down to a level that was inaudible. It has been restored to the original level and the item is closed.

2) MKS 4611

During ACIS V&V, two red light shades on two emergency off switches were found to be broken.

They are:

EOS0007-01

EOS0005-01

The shades have been replaced and the item closed.

All checklists and the non-conformance forms are attached.

Micro84 checks were undertaken separately by HSE staff in February and again in August of 2008. No problems were encountered. Checklists for the Micro84 ACIS V&V are kept in a binder in room 1021.

5.0 REFERENCE MATERIALS

[1] Personnel Safety Lockup System-West 1.2.37.2.Rev.0

[2] Personnel Safety Lockup System Micro84 Program-West 1.2.37.3.Rev.0

[3] LINAC-LTB1 Lockup Procedure" 2.7.37.2 Rev.2

6.0 ATTACHMENTS

LINAC Validation and Verification Checklists

Non-conformance reports (MKS 4611, 4612)

ATTACHMENTS FOR

**1.5.52.5 Rev. 0 – LINAC Access Control Interlock System Validation
and Verification Report – Oct. 2008**

ROUTINE #1

EMERGENCY OFF AND PERSONNEL SAFETY SYSTEM

Test as follows:

- One person in control room and one person in the room under test.
- The person in the room under test will press and reset each panic station checking for the following:
 - a) proper operation of station light.
 - b) proper operation of button locking device.
- The person in the control room will observe R1021.3-01, P1021.3-03 (P296) for the following:
 - a) proper operation of the horn.
 - b) proper operation of the indicating light.
- All faults must be recorded in maintenance log.

[illegible]

ROUTINE #1-1

PERSONNEL SAFETY INTERLOCK SYSTEM

Check each station and area lock-up light (see Figure #1 in the Personnel Inspection Lock-Up Procedure).

Zone #1, Area 0002

Station #1	✓	✓	✓							
Lock-up indicator	✓	✓	✓							

Zone #1, Areas 0001/0010/0011/00S2/10S2

Station #2	✓	✓	✓							
Station #3	✓	✓	✓							
Station #4	✓	✓	✓							
Station #5	✓	✓	✓							
Door (D2) Indicator	✓	✓	✓							
Flashing Light	✓	✓	✓							

Date:

23 MAY 08/001

Initials:

PLR LL LL
E

4 966-8535
1 966-8532

Zone #2, Area 0008

Electrical cover off

Station #1	✓	✓	✓							
Lock-up Indicator	✓		✓							

Zone#2, Areas 0009/00C1/0007

Station #2	✓	✓	✓							
Station #3	✓	✓	✓							
Station #4	✓	✓	✓							
Station #5	✓	✓	✓							
Station #6	✓	✓	✓							
Lock-up Indicator	✓	✓	✓							
Flashing Light	✓	✓	✓							

~~lights~~
~~replacement~~

Date: 23 May 2001

Initials: PK LE

Zone#3, Areas 0003/01S6

Station #1	✓	✓	✓							
Station #2	✓	✓	✓							
Station #3	✓	✓	✓							
Station #4	✓	✓	✓							
Station #5	✓	✓	✓							
Lock-up Indicator	✓	✓	✓							

Zone#3, Areas 0014/0012/0013/0103

Station #6	✓	✓	✓							
Station #7	✓	✓	✓							
Station #8	✓	✓	✓							
Station #9	✓	✓	✓							
Station #10	✓	✓	✓							
Station #11	✓	✓	✓							
Station #12	✓	✓	✓							
Lock-up Indicator	✓	✓	✓							

Zone#3, Areas 0102/1007

Station #13	✓	✓	✓							
Station #14	✓	✓	✓							
Station #15	✓	✓	✓							
Station #16	✓	✓	✓							
Lock-up Indicator	✓	✓	✓							
Flashing Light	✓	✓	✓							

Date: 22 MAY 06001

Initials: Rb Le

Zone#4, Areas 0004/0004.1

Station #1	✓	✓								
Station #2	✓	✓								
Station #3	✓	✓								
Station #4	✓	✓								
Lock-up Indicator	✓	✓								

Zone #4, Area 0006

Station #5	✓	✓								
Station #6	✓	✓								
Lock-up Indicator	✓	✓								

Zone #4, Area 0005

Station #7	✓	✓								
Lock-up Indicator	✓	✓								

Zone#4, Areas 00S1/01S1/10S1

Station #8	✓	✓								
Station #9	✓	✓								
Station #10	✓	✓								
Lock-up Indicator	✓	✓								
Flashing Light	✓	✓								

Date:

23 MAY 061001

Initials:

PK LC
SC

Zone #5, Areas 0108/1300

Station #1	✓		✓							
Station #1	✓		✓							
Lockup-Indicator	✓		✓							
Flashing Light	✓		✓							

Date: 23 MAY 0800Z _____

Initials: JS HL _____
E

CONTROL ROOM STATUS PANEL LIGHTS: Rack R1021.3-01 Panel P1021.3-09

ECS	✓		✓							
LINAC Zone #1	✓		✓							
South Access Storage Hut	✓		✓							
South Access	✓		✓							
Crane well Zone #2	✓		✓							
EA2 Rear Storage	✓		✓							
EA2 Rear Access	✓		✓							
EA2	✓		✓							
Switchyard	✓		✓							
Power Supply Room	✓		✓							
EA3	✓		✓							
EA3 Access	✓		✓							
EA3 Access Stairway Zone #4	✓		✓							
BLTB Tunnel Zone #5	✓		✓							
	✓									
All Zones	✓		✓							

Horns

Zones #1 Thru #5	✓		✓							
------------------	---	--	---	--	--	--	--	--	--	--

Date: 23 May 081002

Initials: R/L L

ROUTINE #2

PERSONNEL SAFETY LOCK-UP

1. Turn off two single pole breakers (Room 1021, P-1U, CCT #9,11).
Connect test lead to TB1-7 (test lead is 18" #14 wire with alligator clip on one end).
Reset aforementioned circuit breaker.
2. Apply power to Flasher #1 at terminal #2. Flasher for zone #1 is now active.
Check lights for proper operation.
Repeat for Flasher #2 zone #2, Flasher #3 zone #3, Flasher #4 zone #4, and
Flasher #5 zone #5.
3. Remove the 4 wires at the bottom of TB1-25.
Apply power to each wire checking for proper operation of horns as listed below:

Wire #25 - 1 - South Access, EA1, Linac Hall, EA3 Access
Wire #25 - 2 - EA3
Wire #25 - 3 - Power Supply Room
Wire #25 - 4 - Switchyard Room
Wire #25 - 5 - EA2
4. Turn off breakers from Step #1. Remove test lead; reconnect wires to TB1-25.
Reset breaker.

Date of Maintenance	Technician's Initials
May 24	42 30/4
05/10/02	LC

* Horn in
Stairwell 1
above LVS #9
is not functioning

MKS 4612

ROUTINE #2-1

PERSONNEL SAFETY INTERLOCK SYSTEM

ZONE #1: Door or Gate Switches and Locks

Zone #1/Zone #3 Emg. Gate #1	✓	✓								
Area 0002/0001 West Wall Gate #2	✓	✓								
Area 0002/0001 North Wall Gate #3	✓	✓								
Zone #1/Zone #3 Radiation Door #1	✓	✓								
Zone #1/Zone #2 Door #1	✓	✓								
Zone #1/Zone #2 Emg. Exit Gate #4	✓	✓								
Area 0010/00S2 Door #2	✓	✓								
Area 10S2/1008 Door #4	✓	✓								
Area 10S2 Emg. Exit Door #3	✓	✓								

2

Date:

MAY 23/01 0700

Initials:

RL LL
SR

Zone #2: Door or Gate Switches and Locks

Zone #2/Zone #1 Emg. Exit Gate #4	✓	✓								
Zone #2/Zone #1 Door #1	✓	✓								
Area 0008/0009 Gate #6	✓	✓								
Zone #2/Zone #3 Gate #7	✓	✓								
Zone #2/Zone #4 Gate #8	✓	✓								
Area 00C1/0009 Door #5	~	✓								
Crane Stair Access Door #12	✓	✓								

1 1

Zone #3: Door or Gate Switches and Locks

Zone #3/Zone #1 Emg. Exit Gate #1	✓	✓								
Zone #3/Zone #1 Radiation Door #1	✓	✓								
Zone #3/Zone #2 Gate #7	✓	✓								
Zone #3/Zone #3 Gate #11	✓	✓								
Area 0003/0013 Gate #15	✓	✓								
Zone #3/Zone #4 Door #6	✓	✓								
Zone #3/Zone #4 Gate #9	✓	✓								
Area 0012/0014 Door #7	✓	✓								
Area 0012/0013 Gate #13	✓	✓								
Area 0012/0013 Gate #14	✓	✓								
Area 0103/0102 Door #12	✓	✓								
Area 0102/01S6 Door #9	✓	✓								
Area 1006/1007 Door #10	✓	✓								

Date:

May 23/21 08/1002

Initials:

Mr. El
E

Zone #4: Door or Gate Switches and Locks

Zone #4/Zone #3 Gate #9	✓		✓							
Zone #4/Zone #3 Door #6	✓		✓							
Zone #4/Zone #5 Gate #16	✓		✓							
Zone #4/Zone #3 Gate #11	✓	✓	✓							
Area 0004/0006 Gate #10	✓	✓	✓							
Zone #4/Zone #2 Gate #8	✓	✓	✓							
Area 0006/00S1 Door #8	✓	✓	✓							
Area 00S1/0005 Door #11	✓	✓	✓							
Area 01S1/0104 Door #17	✓	✓	✓							
Area 10S1 Gate #6	✓	✓	✓							

Zone #5: Door or Gate Switches and Locks

Zone #5/Zone #4 Gate #16	✓	✓	✓							
Area 0108/1300 Gate #18	✓		✓							

Gate 21

✓

Gate 19

✓

Gate 20

✓

Date:

May 23/01

D81002

Initials:

RP
SE

LC

LINAC V&V CHECKLIST

EOS

red cover broken

1	12	23	34
2	13	24	35
3	14	25	36
4	15	26	37
5	16	27	38
6	17	28	39
7	18	29	40
8	19	30	41
9	20	31	42
10	21	32	43
11	22	33	44

red cover is broken

MKS 4611

~~MKS 4612~~

Problem Report 4611

Fields

Summary LINAC V&V Broken red light shades on EOS0007-01 & EOS0005-01

State Closed

Priority Low

Install Schedule

Impact Low/Medium

Next Review

Project /11 HSE/ACIS/Accelerators

Assigned User Lyndon Cowles (cowlesl)

Assigned Group

Description Please replace these broken items

Analysis Design Use WO 07-00294
Return to Lyndon to verify work

Implementation shades replaced

ECR Raised false

Forward Relationships

ID	Type	State	Summary	Assigned User	Created By	Order	Relationship	Flags
----	------	-------	---------	---------------	------------	-------	--------------	-------

Required by	2-Oct-2008
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Percent Complete	100
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Work Package

ID	Type	State	Summary	Assigned User	Created By	Order	Relationship	Flags
----	------	-------	---------	---------------	------------	-------	--------------	-------

Work Order

ID	Type	State	Summary	Created By	Order	Relationship Flags
Estimated Man Days		0.5				

Relationships

Backward Relationships

ID	Type	State	Summary	Assigned User	Created By	Order	Relationship Flags
----	------	-------	---------	---------------	------------	-------	--------------------

Attachments

Attachments

Problem Report 4612

Fields

Summary	LINAC V&V Horn is not working
State	Closed
Priority	Low
Install Schedule	
Impact	Low/Medium
Next Review	
Project	/11 HSE/ACIS/Accelerators
Assigned User	Lyndon Cowles (cowlesl)
Assigned Group	
Description	The claxon in stairwell 1 zone 4, above LUS 9 is not working. Please repair.
Analysis Design	Use WO 07-00294 Return to Lyndon to verify work
Implementation	bench tested horn, it worked but is turned down because it is in the stairway applied power to wire 1, as in v&v, horn worked in place
ECR Raised	false

Forward Relationships

ID	Type	State	Summary	Assigned User	Created By	Order	Relationship	Flags
----	------	-------	---------	---------------	------------	-------	--------------	-------

Required by	2-Oct-2008
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Percent Complete	100
------------------	-----

Work Package

ID	Type	State	Summary	Assigned User	Created By	Order	Relationship	Flags
----	------	-------	---------	---------------	------------	-------	--------------	-------

Work Order

ID	Type	State	Summary	Created By	Order	Relationship Flags
Estimated Man Days						0.5

Relationships

Backward Relationships

ID	Type	State	Summary	Assigned User	Created By	Order	Relationship Flags
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Attachments

Attachments